BUMPER SUB

FISHING BUMPER SUB

JARRING

GENERAL DESCRIPTION

DURABLE FISHING BUMPER SUB MADE OF HEAT-TREATED ALLOY STEEL FOR SUSTAINED BUMPING WITH MINIMAL MAINTENANCE. RUN IN THE STRING JUST ABOVE THE FISHING TOOL OR SAFETY JOINT, IT LETS THE OPERATOR DELIVER SOLID DOWNWARD OR UPWARD BLOWS AT WILL, FOR EXAMPLE TO RELEASE A FISHING TOOL THAT CANNOT PULL THE FISH. A HEXAGONAL MANDREL SLIDES IN A MATCHING BODY, SO TORQUE IS TRANSMITTED OVER THE WHOLE STROKE. IT CAN ALSO ACT AS A FEED-OFF TOOL FOR INTERNAL PIPE CUTTING.

TECHNICAL CHARACTERISTICS

O.D.	I.D.	Tensile at yield (lb)	Torque at yield (ft-lb)
1-13/16	3/8	68,720	1,020
1-29/32	3/8	78,620	850
2-1/4	1/2	172,900	2,440
2-1/2	1/2	160,600	1,940
3-1/8	1-1/2	158,600	3,300
3-1/8	1	224,900	4,200
3-3/8	1	256,000	5,740
3-3/4	1-1/2	316,200	6,000
3-3/4	1-1/4	268,500	9,400
3-3/4	1-7/8	231,000	6,810
4-1/8	1-1/2	370,200	9,270
4-1/4	1-15/16	305,000	9,200
4-1/4	1-1/2	439,000	9,160
4-1/2	2-3/8	336,600	10,240

O.D.	I.D.	Tensile at yield (lb)	Torque at yield (ft-lb)
4-5/8	1-7/8	442,000	13,440
4-5/8	2	465,900	12,280
4-3/4	1-1/2	581,100	13,800
4-3/4	2	465,900	8,200
5-1/2	2	613,400	22,070
5-3/4	2	723,600	26,850
6	2	767,000	29,490
6-1/8	2-1/4	768,600	30,840
6-1/4	2-1/4	864,400	38,000
6-1/2	2-1/4	864,400	42,000
6-3/4	2-3/4	871,000	37,600
7-3/4	3-1/2	1,060,000	50,400
8	3-1/2	1,060,000	54,600
9	4	1,600,000	64,110

Dimensions in inches. Standard stroke 20 in; strokes of 30, 36, 42, 48 and 60 in also available (convert by changing mandrel and middle body only). Tensile strengths are calculated theoretical yield values (about +/-20%); same values typically apply to other strokes. Figures are a guide only: apply a safety factor.

DESIGN FEATURES

- **Solid blows both ways:** downward on closing, upward on opening against the knocker.
- **Hexagonal mandrel:** torque transmitted over the full stroke.
- **Alloy steel body:** heat-treated parts for bumping, tension and torque.
- **Sealed knocker:** packing rings seal mandrel to middle body; optional chevron packing.
- **Low maintenance:** wash and lubricate after use; check seals for wear.

OPERATING NOTES

- **Bump down:** pull to fully open plus stretch, then drop the string and brake before the sub closes for repeated blows, or drop freely for one solid blow.
- **Jar up:** take moderate stretch, drop by that stretch only and brake; the sub rebounds open and strikes the knocker.
- **At surface:** hold a sledge between body and mandrel shoulder, slack off and pull it out to release a fishing tool.

HOW TO ORDER

Please specify: 1. size (O.D. x I.D.) 2. connection size and type 3. length of stroke 4. recommended spares: 8 non-extrusion rings, 8 seal protector rings, 16 knocker seals.